

INTRODUCTION TO EMBEDDED SYSTEMS & IOT DEVICES DESIGN

REPORT TASK-1

PART 1 - INTRODUCTION

What is an Embedded System?

An embedded system is a computer system designed to perform a specific function within a larger system. Unlike a general-purpose computer, it is programmed for one dedicated task and often operates in real time.

An embedded system consists of:

- Hardware (Microcontroller, Sensors, Actuators)
- Software (Embedded Program)
- Input and Output Devices

Features

- Performs a dedicated task
 - Low power consumption
 - High reliability
 - Compact size
 - Real-time operation
 - Cost-effective
-

Real-World Applications

Smart Home

- Smart lighting
- Smart door locks
- Home automation

Automotive

- Airbag system
- ABS braking
- Engine Control Unit (ECU)

Healthcare

- ECG machine
- Digital thermometer
- Blood pressure monitor

Robotics

- Industrial robots
- Line follower robots
- Autonomous vehicles

Consumer Electronics

- Washing machine
- Microwave oven

Difference Between Microcontroller and Microprocessor

Microcontroller

CPU + RAM + ROM on single chip

Low power consumption

Low cost

Used in embedded systems

Performs dedicated tasks

Example: Arduino UNO

Microprocessor

Only CPU

High power consumption

Expensive

Used in computers

Performs multiple tasks

Example: Intel Core i5

PART 2 - COMPONENT STUDY

Arduino UNO

Arduino UNO is an open-source microcontroller board based on the ATmega328P.

Specifications

- Operating Voltage: 5V
- Digital Pins: 14
- Analog Pins: 6
- Flash Memory: 32 KB
- Clock Speed: 16 MHz

Applications

- Home Automation
- Robotics
- IoT Projects

ESP32

ESP32 is a powerful Wi-Fi and Bluetooth-enabled microcontroller.

Features

- Built-in Wi-Fi
- Bluetooth
- Dual-core processor

Applications

- IoT & Wireless Monitoring

Raspberry Pi Pico

Raspberry Pi Pico uses the RP2040 microcontroller.

Features

- Dual-core ARM Cortex-M0+
- 264 KB RAM
- Low power
- Programmable using C/C++ and MicroPython

Applications

- Robotics
- Embedded Systems
- Automation

Sensors

PIR Motion Sensor

A PIR (Passive Infrared) sensor detects movement by sensing infrared radiation emitted from the human body.

Working

- Detects body heat
- Sends HIGH signal to Arduino
- Arduino activates LED/Buzzer

Applications

- Security systems
 - Automatic doors
 - Smart lighting
-

Temperature Sensor (LM35)

Measures surrounding temperature.

Applications

- Weather monitoring
 - Medical devices
 - Industrial automation
-

Light Sensor (LDR)

Measures light intensity.

Applications

- Automatic street lights
- Smart lighting
- Solar tracking

Mini Project

Project Title

Motion Detection Alarm Using Arduino UNO

Objective

To detect human motion using a PIR sensor and activate an LED whenever motion is detected.

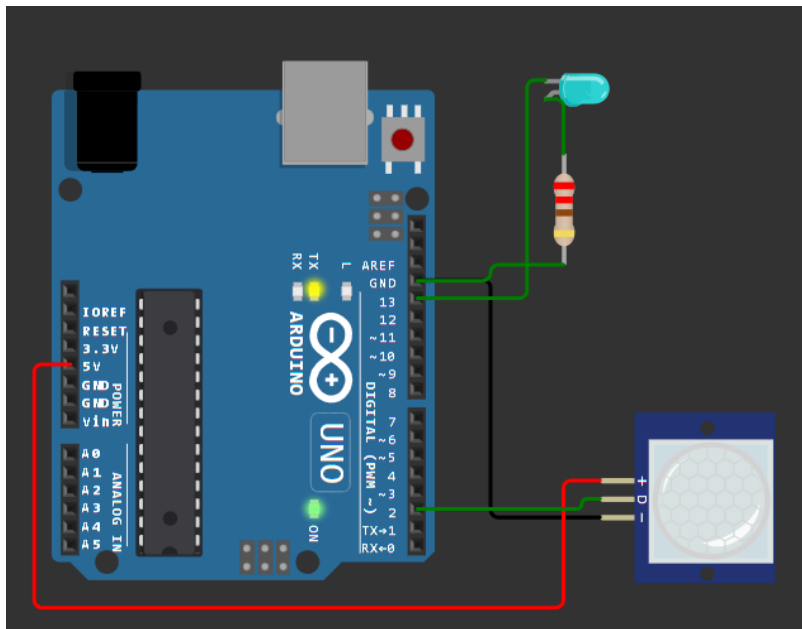
Components Required

- Arduino UNO
- PIR Motion Sensor
- LED
- 220 Ω Resistor
- Breadboard
- Jumper Wires

Circuit Connections

<u>Component</u>	<u>Arduino Pin</u>
PIR VCC	5V
PIR GND	GND
PIR OUT	Digital Pin 2
LED Positive	Digital Pin 13
LED Negative	220Ω Resistor → GND

Circuit Diagram



ARDUINO CODE

```
1  const int pirPin = 2;
2  const int ledPin = 13;
3
4  void setup()
5  {
6      pinMode(pirPin, INPUT);
7      pinMode(ledPin, OUTPUT);
8
9      Serial.begin(9600);
10 }
11
12 void loop()
13 {
14     int motion = digitalRead(pirPin);
15
16     if(motion == HIGH)
17     {
18         digitalWrite(ledPin, HIGH);
19         Serial.println("Motion Detected!");
20     }
21     else
22     {
23         digitalWrite(ledPin, LOW);
24         Serial.println("No Motion");
25     }
26
27     delay(500);
28 }
```

Working Principle

1. PIR sensor continuously monitors nearby movement.
2. When a person enters its detection range, the sensor outputs a HIGH signal.
3. Arduino reads the signal from pin D2.
4. Arduino turns ON the LED connected to pin D13.
5. When no movement is detected, the LED turns OFF.

Expected Output

- **Motion detected:** LED ON and Serial Monitor displays "Motion Detected!"
- **No motion:** LED OFF and Serial Monitor displays "No Motion"

Real-World Applications

- Home security systems
- Office security
- Automatic room lighting
- Smart street lights

Conclusion

The Motion Detection Alarm demonstrates the basic concepts of embedded systems by interfacing a PIR sensor with an Arduino UNO. It shows how sensors, microcontrollers, and output devices work together to create an intelligent system. This project is simple, cost-effective, and has practical applications in home automation and security.